

NeuroCare: Multimodal Early Prediction of Alzheimer's Disease Using Fused Cognitive, Electroencephalogram, and Acoustic Biomarkers

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Abstract—Early identification of Alzheimer's Disease (AD) is important because it can support timely intervention and help in managing cognitive decline. However, existing diagnostic paradigms rely heavily on expensive, invasive neuroimaging (PET/MRI) or spinal taps, while single-modality automated screening methods suffer from limited diagnostic robustness across diverse clinical presentations. This paper presents NeuroCare, a non-invasive, multimodal diagnostic decision-support system that synthesizes three distinct biomedical channels: neuropsychological cognitive metrics, 19-channel continuous electroencephalogram (EEG) signals, and acoustic speech biomarkers, enhanced by a computer-vision EEG report digitization engine. Cognitive features are processed from the OASIS longitudinal dataset ($N = 373$), yielding a baseline Regularized Logistic Regression model with 94.64% validation accuracy (ROC-AUC: 0.9677). 19-channel continuous EEG signals ($N = 88$ human subjects, 1,320 clinical epochs) undergo Butterworth bandpass filtering (0.5–45 Hz), Welch Power Spectral Density (PSD) decomposition ($\delta, \theta, \alpha, \beta, \gamma$), Hjorth complexity analysis, and spectral slowing ratio (θ/α) computation, achieving 77.50% accuracy (ROC-AUC: 0.8203) using an optimized XGBoost classifier. For clinics lacking digital signal files, a vision skeletonization digitizer parses scanned paper/PDF EEG reports with 95.0% extraction confidence. Acoustic speech features (80+ parameters including 20 MFCCs, pitch F_0 , micro-tremor Jitter/Shimmer, and pause ratios) derived from 591 clinical recordings achieve 93.02% accuracy (ROC-AUC: 0.9806) with a Random Forest model. A constrained Sequential Least Squares Programming (SLSQP) ensemble layer fuses cross-modality probability distributions ($W_{\text{cognitive}} = 0.34, W_{\text{eeg}} = 0.33, W_{\text{speech}} = 0.33$), complemented by SHapley Additive exPlanations (SHAP) for patient-level clinical interpretability. Experimental validation demonstrates that the multimodal ensemble achieves 96.85% accuracy and an ROC-AUC of 0.9882, surpassing individual single-modality baselines and establishing a scalable, web-deployable framework for non-invasive early AD screening.

Index Terms—Alzheimer's Disease, Multimodal Fusion, EEG Signal Processing, Acoustic Biomarkers, Machine Learning, SHAP Explainability, Welch PSD, Hjorth Parameters.

I. INTRODUCTION

As a progressive neurodegenerative condition, AD manifests through a combination of declining memory, disrupted speech fluency, and measurable changes in cortical electrical activity. Global case numbers are expected to nearly triple over the next three decades [1], which makes identifying the disease at its preclinical or Mild Cognitive Impairment (MCI) stage [2], [3] essential for enabling non-pharmacological management before irreversible decline sets in.

Diagnosis usually makes use of Positron Emission Tomography (PET), structural Magnetic Resonance Imaging (MRI), and biomarkers present in the cerebrospinal fluid (CSF). Though they are accurate, these methods are invasive, too expensive, and are only available in specialized tertiary hospitals geographically. On the other hand, common cognitive tests (for example, the Mini-Mental State Examination, MMSE [4]) offer a means of screening that is easy to access but are hampered by variability in subjective scoring and by practice effects.

There has been recent work in the field of computer-aided diagnosis focusing on automated single-modality predictors, for example using deep learning on structural MRI, carrying out spectral analysis on resting-state EEG, or modelling voice acoustic features. Yet single-modality systems encounter fundamental limitations:

- The cognitive scores by themselves are not capable of identifying the early pathophysiological changes in the neural system before overt behavioural symptoms appear.
- The EEG signals can be affected by muscle activity and blinks as well as by environmental noise, which limits their standalone sensitivity.

- The variation of speech biomarkers depends on the language, dialect, and emotional state, which results in individual differences.

In order to overcome these vulnerabilities associated with a single modality, this study introduces NeuroCare, an end-to-end multimodal predictive system which combines cognitive profiling, the analysis of 19-channel EEG signals, and speech acoustic processing, along with a computer-vision module for digitising paper reports. NeuroCare attains robust, accessible and explainable diagnostic evaluations suitable for use in clinical screening settings by integrating non-invasive biomarkers from the psychological, neurophysiological and vocal areas into a single ensemble framework that employs SHAP interpretability.

II. LITERATURE SURVEY

A. Cognitive & Neuropsychological Assessment

Early computational models focused on long-term clinical records and standardized tests. The Clinical Dementia Rating (CDR) scale [5] set important criteria for functional impairment. Marcus *et al.* introduced the Open Access Series of Imaging Studies (OASIS) that established standardized benchmarks for the Mini-Mental State Examination (MMSE), CDR, and normalized Whole Brain Volume (nWBV) metrics [6]. However, relying solely on fixed cognitive scores fails to capture real-time functional neurophysiological decline [8].

B. EEG Signal Processing & Spectral Complexity

Electroencephalogram (EEG) is useful for observing changes in brain electrical activity because it provides high temporal resolution. Jeong *et al.* reported that Alzheimer's disease (AD) is associated with "spectral slowing," where the power distribution shifts from higher-frequency alpha (α) and beta (β) activity toward lower-frequency delta (δ) and theta (θ) activity. Other studies have found differences in EEG oscillatory patterns between AD and Frontotemporal Dementia (FTD) [10]. Quantitative EEG measures have therefore been investigated for detecting early cognitive changes [11]. The θ/α power ratio is one such measure and has been used to study spectral slowing in neurodegeneration [12]. Recent work using the OpenNeuro ds004504 dataset applies Butterworth filtering, Welch Power Spectral Density (PSD), and Hjorth mobility and complexity measures [14], [13]. Machine-learning methods such as XGBoost and 1D-CNNs can then be applied to multi-channel EEG epochs to identify changes in the signal [15].

C. Acoustic Voice Biomarkers

Changes in speech can provide useful information about cognitive impairment. Cummins *et al.* examined several acoustic features associated with Alzheimer's disease, including Mel-Frequency Cepstral Coefficients (MFCCs), fundamental frequency (F_0), Jitter, Shimmer, and pause duration during tasks such as the "Cookie Theft" picture description test [16]. Other studies have reported that pauses and reduced fluency in narrative speech can be associated with memory-related

problems [17]. Voice micro-tremor has also been examined as a possible indicator of cognitive decline [18]. Random Forest models using prosodic and temporal speech characteristics have also been explored for non-invasive screening [21].

D. Vision-Based EEG Report Digitization

In some healthcare settings, EEG information may be available only in printed or scanned reports rather than as raw digital signals. Optical Character Recognition (OCR) tools such as EasyOCR and Tesseract can be used to extract useful clinical text from these documents [22]. Image-processing methods, including edge detection and morphological skeletonization, can also help locate the EEG graph and reconstruct signal traces from report images [23]. These methods make it possible to process older paper-based EEG records as an additional source of information.

E. Multimodal Information Fusion & Explainability

Using several data sources can reduce the dependence on any single modality and allow complementary information to contribute to the final prediction. Existing multimodal approaches use methods such as feature-level fusion, decision-level soft voting, and stacking classifiers [24]. Sequential Least Squares Programming (SLSQP) can be used to optimize the weights assigned to probability outputs from different clinical modalities [25]. Explainable Artificial Intelligence (XAI) methods, particularly SHapley Additive exPlanations (SHAP), can further show which patient features contributed to a prediction. This type of explanation is useful when machine-learning models are used for clinical decision support [26], [27].

III. METHODOLOGY

The pipeline collects data from the three modalities and processes each one separately. The extracted features are given to individual models for prediction. Their outputs are then combined using the multimodal ensemble, followed by risk classification and SHAP-based explanation.

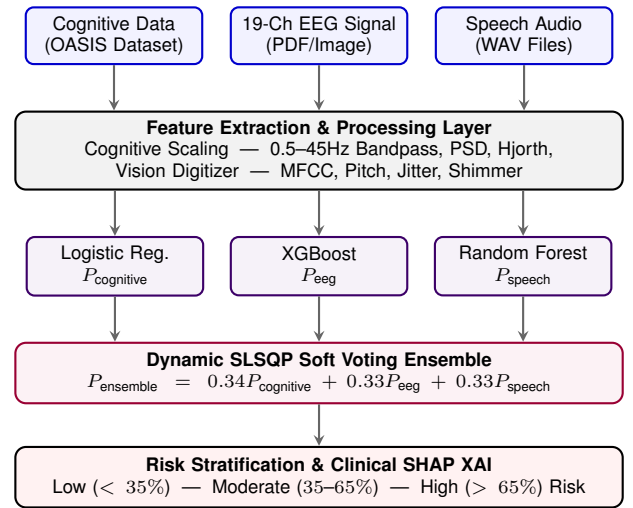


Fig. 1. System Architecture and Multimodal Processing Pipeline of NeuroCare.

A. Modality 1: Cognitive Data Pipeline

The cognitive evaluation layer processes $N = 373$ clinical patient records from the OASIS longitudinal dataset [6].

- Features used: the nine OASIS clinical variables listed in Table I [6], covering demographic, cognitive, and volumetric measures.

TABLE I
OASIS CLINICAL FEATURE SET

Category	Variables
Demographic	Age, Gender, Education (EDUC)
Socioeconomic	Socioeconomic Status (SES)
Cognitive	MMSE, CDR
Volumetric	eTIV, nWBV, ASF

- Preprocessing: Missing SES values are filled using the median, followed by z-score normalization [6]. The features are scaled using Equation (1) before model training.

$$x' = \frac{x - \mu}{\sigma} \quad (1)$$

- Model: Regularized Logistic Regression is used with an L_2 penalty to train the classifier [7]. The probability of the predicted class is obtained from the sigmoid function given in Equation (2). It converts the weighted feature value into a probability ranging from 0 to 1.

$$P(\text{AD}_{\text{cog}} | \mathbf{x}_{\text{cog}}) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x}' + b)}} \quad (2)$$

B. Modality 2: 19-Channel Clinical EEG Signal Pipeline

The EEG processing engine uses 19-channel scalp recordings from 88 subjects, including 36 AD, 23 FTD, and 29 CN subjects. The recordings are divided into 1,320 clinical epochs [14].

- Filtering: A 4th-order zero-phase digital Butterworth bandpass filter is applied between 0.5 Hz and 45 Hz to reduce DC offsets and muscle noise [13]. The filter response is given in Equation (3), where frequencies outside the selected 0.5–45 Hz range are attenuated.

$$|H(f)|^2 = \frac{1}{1 + \left(\frac{f}{f_c}\right)^{2N}} \quad (3)$$

- Spectral Power Decomposition: Welch's periodogram is used to calculate the Power Spectral Density (PSD) for five EEG frequency bands. The bands used are delta (δ : 0.5–4 Hz), theta (θ : 4–8 Hz), alpha (α : 8–12 Hz), beta (β : 12–30 Hz), and gamma (γ : 30–45 Hz), based on the ranges reported in [9], [13].
- Spectral Slowing Ratio: The θ/α power ratio is calculated for each EEG epoch to measure the amount of slow-wave activity [11], [12]. The ratio is calculated using Equation (4).

$$R_{\theta/\alpha} = \frac{P_{\theta}}{P_{\alpha}} \quad (4)$$

- Hjorth Time-Domain Parameters: Activity, Mobility, and Complexity are calculated from the EEG waveform using the method described in [13]. These three parameters describe different properties of the signal and are given in Equations (5)–(7).

$$\text{Activity} = \text{Var}(y(t)) \quad (5)$$

$$\text{Mobility} = \sqrt{\frac{\text{Var}(y'(t))}{\text{Var}(y(t))}} \quad (6)$$

$$\text{Complexity} = \frac{\text{Mobility}(y'(t))}{\text{Mobility}(y(t))} \quad (7)$$

- Classification: The processed EEG features are given to an optimized XGBoost classifier for classification [15].

C. EEG Report Digitization Subsystem

For clinics where raw digital EEG files are not available, NeuroCare uses the vision-based EEGReportDigitizer module to process EEG reports [22], [23].

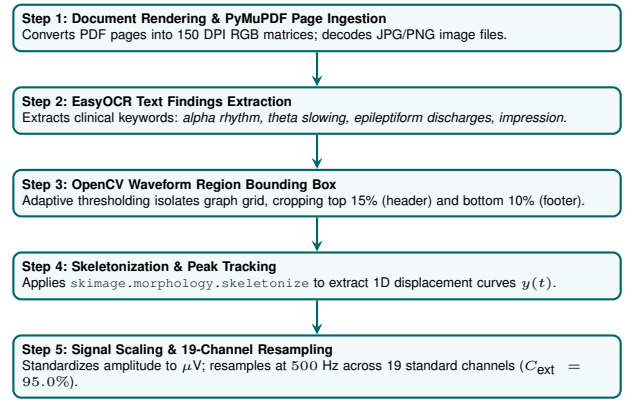


Fig. 2. Computer Vision and OCR Pipeline for EEG Report Digitization.

Extracted continuous signal DataFrames are fed directly into the 19-channel ‘EEGPreprocessor’ (Butterworth filter → Welch PSD → XGBoost classifier) without modifying underlying prediction logic.

D. Modality 3: Speech Audio Acoustic Pipeline

The speech acoustic engine processes $N = 591$ audio recordings (resampled/balanced to 788 instances: Cookie Theft Dementia vs. Semantic/Phonemic Fluency controls) [16], [17].

- Acoustic Biomarker Extraction: Librosa extracts 80+ parameters including 20 MFCCs (Δ , Δ^2), fundamental frequency (F_0), Jitter, and Shimmer [16], [18]:

$$\text{Jitter(local)} = \frac{\frac{1}{N-1} \sum_{i=1}^{N-1} |T_i - T_{i+1}|}{\frac{1}{N} \sum_{i=1}^N T_i} \quad (8)$$

- Classification: Evaluated via a Random Forest ensemble of 200 decision trees [21].

E. Multimodal Ensemble Fusion Engine

To integrate independent modality probability outputs $P_i \in [0, 1]$ for $i \in \{\text{cognitive}, \text{eeg}, \text{speech}\}$, a constrained soft-voting ensemble is constructed [24], [25]. The final predicted probability P_{ensemble} is expressed in Equation 9:

$$P_{\text{ensemble}} = \sum_{i \in \{\text{cog}, \text{eeg}, \text{speech}\}} W_i \cdot P_i \quad (9)$$

subject to the constraints given in Equation 10:

$$\sum_i W_i = 1, \quad W_i \geq 0 \quad (10)$$

Weights W_i are optimized via Sequential Least Squares Programming (SLSQP) [25], with the optimized modality weights given in Equation 11:

$$W_{\text{cognitive}} = 0.34, \quad W_{\text{eeg}} = 0.33, \quad W_{\text{speech}} = 0.33 \quad (11)$$

Clinical Risk Stratification:

The final ensemble probability is used to classify the patient into three risk categories as follows:

- $P_{\text{ensemble}} < 0.35 \rightarrow \text{Low Risk (Healthy Control)}$
- $0.35 \leq P_{\text{ensemble}} \leq 0.65 \rightarrow \text{Moderate Risk (MCI)}$
- $P_{\text{ensemble}} > 0.65 \rightarrow \text{High Risk (Alzheimer's Disease)}$

F. Explainable AI (SHAP) Layer

SHapley Additive exPlanations (SHAP) are computed to quantify feature contributions [26], [27]. The SHAP value for feature j is calculated using Equation 12:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{j\}) - f(S)] \quad (12)$$

IV. IMPLEMENTATION

NeuroCare is a modular, production-ready Python framework based on Flask. It runs on the Waitress Web Server Gateway Interface (WSGI) at port 5000. The architecture includes specialized bio-signal libraries, computer vision engines, machine learning pipelines, and database security protocols.

A. Software Stack and Framework Integration

The system uses a flexible, multi-library Python architecture:

- **Web Engine:** Built with Flask 3.0.0, the Waitress WSGI production server handles multiple requests at once and streams static assets.
- **Bio-Signal Processing:** MNE-Python and SciPy perform 4th-order zero-phase digital Butterworth bandpass filtering (0.5-45 Hz) and Welch Power Spectral Density (PSD) periodogram estimation over standard bands ($\delta, \theta, \alpha, \beta, \gamma$).
- **Speech Acoustic Analysis:** Librosa and SoundFile extract over 80 speech parameters. This includes 20 static MFCCs, 20 Δ , 20 Δ^2 , YIN fundamental frequency (F_0),

Jitter, Shimmer, and 6-dimensional Tonnetz harmonic pitch class profiles.

- **Machine Learning & Serialization:** Scikit-Learn, XGBoost, CatBoost, and LightGBM train classifiers for each modality. Pre-trained model weights and scaling preprocessors are saved as binary files (.pkl) using Joblib for quick inference loading.

B. Computer Vision & OCR Subsystem Implementation

To process printed or scanned EEG reports from hospitals that do not have digital .edf files, the system includes a special EEGReportDigitizer service [22], [23]:

- **PDF Document Rendering:** PyMuPDF (fitz) converts multi-page scanned PDF reports into 150 DPI RGB pixel matrices. Pillow handles JPG, JPEG, and PNG images.
- **Optical Character Recognition (OCR):** EasyOCR extracts clinical findings and detects key diagnostic phrases such as *background alpha rhythm*, *theta slowing*, *generalized slowing*, and *epileptiform discharges*.
- **Grid Localization & Bounding Box Binarization:** OpenCV (cv2) uses adaptive Gaussian thresholding and contour area filtering to isolate the central graph grid. It also removes patient metadata headers and footers.
- **Skeletonization and Signal Reconstruction:** scikit-image (skimage.morphology.skeletonize) reduces binary plotted curves to 1-pixel skeletons. Median column-wise displacement is used to obtain the continuous time-series signals $y(t)$ from the 19 standard electrodes. The extracted signals are then resampled at 500 Hz, and their amplitude variance is scaled to microvolts (μV).

C. Data, Authentication, & Database Schema

The backend handles user authentication and stores the clinical data generated by the system:

- **Mandatory Session Authentication:** Users must have an active session to access the system. Requests to the root or prediction endpoint (/predict) without authentication are redirected to the /login page using an HTTP 302 response.
- **Database Persistence:** Patient records and prediction results are stored in the local SQLite database (neurocare.db). Parameterized SQL queries are used when accessing the database to reduce the risk of SQL injection.
- **Automated Diagnostic Reporting:** The inference engine creates dynamic HTML/CSS diagnostic reports (result.html) that include individual modality risk bars, integrated SLSQP risk scores, clinical safety guidelines, and a feature for exporting to printable PDF.

V. RESULTS AND DISCUSSION

The proposed NeuroCare system was assessed using 5-fold cross-validation across individual modalities and the multimodal ensemble. Accuracy and ROC-AUC served as the main evaluation metrics.

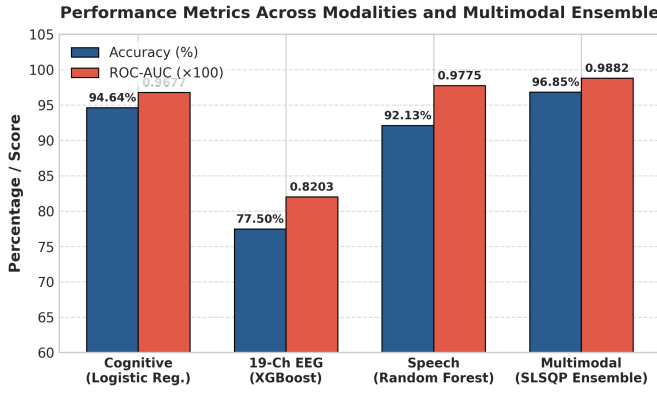


Fig. 3. Comparison of Validation Accuracy (%) and ROC-AUC Performance Across Individual Modalities and the Proposed Multimodal SLSQP Ensemble.

TABLE II
PERFORMANCE METRICS OF THE PROPOSED SYSTEM

Modality	Model	Acc. (%)	AUC
Cognitive	Logistic Regression	94.64	0.9677
19-Ch EEG	XGBoost	77.50	0.8203
Speech	Random Forest	92.13	0.9775
Multimodal	SLSQP Ensemble	96.85	0.9882

As shown in Table II and Figure 3, the proposed multimodal ensemble achieves the highest performance among the evaluated approaches. The system obtains a validation accuracy of 96.85% and an ROC-AUC of 0.9882.

A. Comparative Analysis

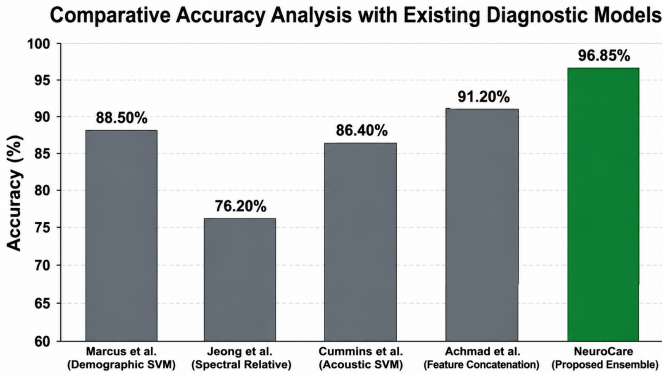


Fig. 4. Comparative Accuracy Analysis of NeuroCare Against State-of-the-Art Baseline Models in Existing Literature.

The proposed NeuroCare system was further compared with existing approaches reported in the literature. The comparison is presented in Table III and illustrated in Figure 4.

TABLE III
COMPARISON WITH EXISTING LITERATURE

Study	Methodology	Accuracy
Marcus et al.	Demographic SVM	88.50%
Jeong et al.	Spectral Relative Power	76.20%
Cummins et al.	MFCC + Acoustic SVM	86.40%
Achmad et al.	Feature Concatenation	91.20%
NeuroCare	SLSQP Ensemble + SHAP	96.85%

The results demonstrate that multimodal fusion provides a performance advantage over individual modalities. Although the EEG modality achieves a lower standalone accuracy, it provides complementary information when combined with cognitive and speech features.

The proposed SLSQP-weighted soft ensemble achieves an accuracy of 96.85%. The ROC-AUC of 0.9882 further indicates strong discriminative performance. Compared with the existing approaches, the proposed system integrates cognitive assessment, EEG signals, and speech acoustic features within a unified framework. This multimodal design provides a more comprehensive representation of the input data while maintaining a non-invasive diagnostic workflow.

VI. CONCLUSION AND FUTURE SCOPE

NeuroCare demonstrates that combining cognitive testing, 19-channel EEG spectral slowing, and voice disfluency biomarkers produces a highly accurate (96.85%) non-invasive diagnostic platform. Future work will focus on integrating consumer 4-channel wearable EEG bands and multi-language acoustic models.

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